

Switch Attention

Towards Dynamic and Fine-grained Hybrid Transformers

Yusheng Zhao¹, Hourun Li¹, Bohan Wu¹, Yichun Yin²,
Lifeng Shang², Jingyang Yuan¹, Meng Zhang², Ming Zhang¹

¹ Peking University / PKU-Anker LLM Lab

² Huawei Technologies Co., Ltd.



Attention cost depends on how far we look

Attention	Context available	Prefill cost
Full attention	All previous tokens	$O(N^2)$
Sliding window	Last w tokens	$O(Nw)$

Static hybrids fix the attention type by layer

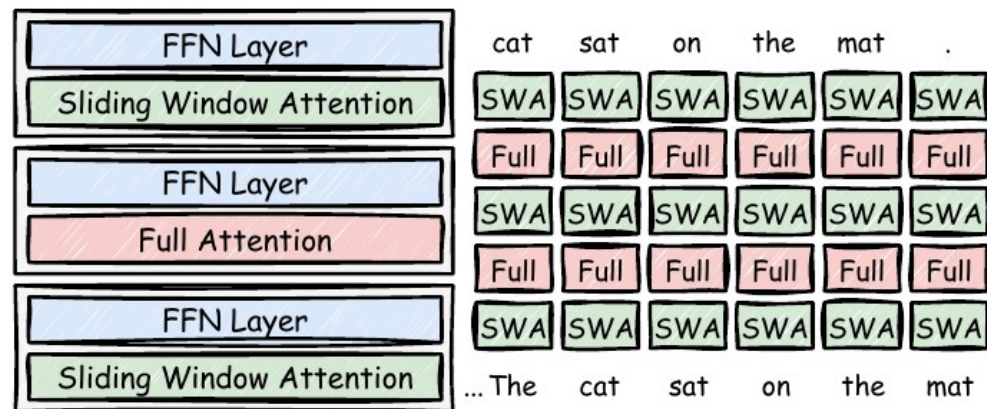
The need for distant context can change from token to token.

A content-dependent choice at every token and layer

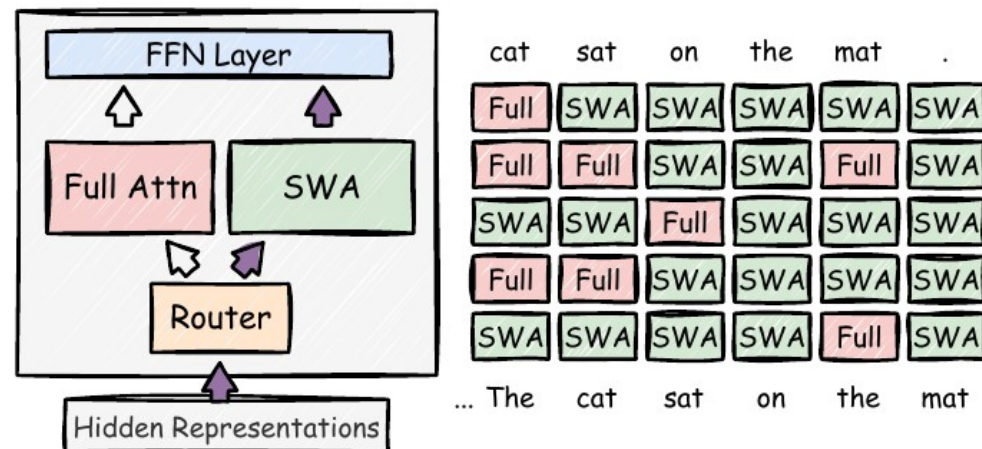
One routing decision

Full attention
for global context

Sliding window
for local context

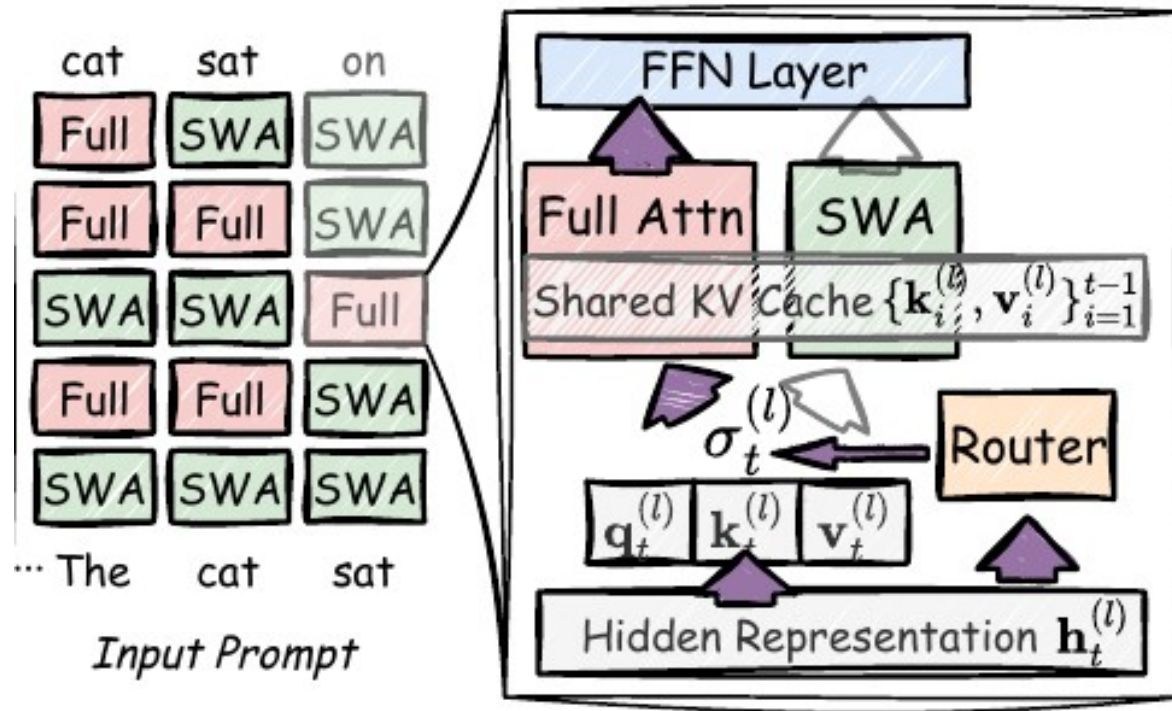


(a) Conventional Methods: Static Hybrid Scheme



(b) Proposed Method: Dynamic and Fine-grained

At inference, the router selects one branch



(b) Decoding Stage

$$p = \text{sigmoid}(R(h))$$

$p > 0.5$: full attention

$p \leq 0.5$: sliding window

Shared QKV projections
One shared KV cache

Only the selected branch runs.

Adaptive regularization encourages local attention

Router penalty = $\gamma \cdot \text{softplus}(R(h))$

$\gamma = \text{base weight} / (\varepsilon + \text{NLL} + \alpha \cdot \text{MSE})$

Easy prediction

Similar branch outputs

Stronger pressure to use SWA

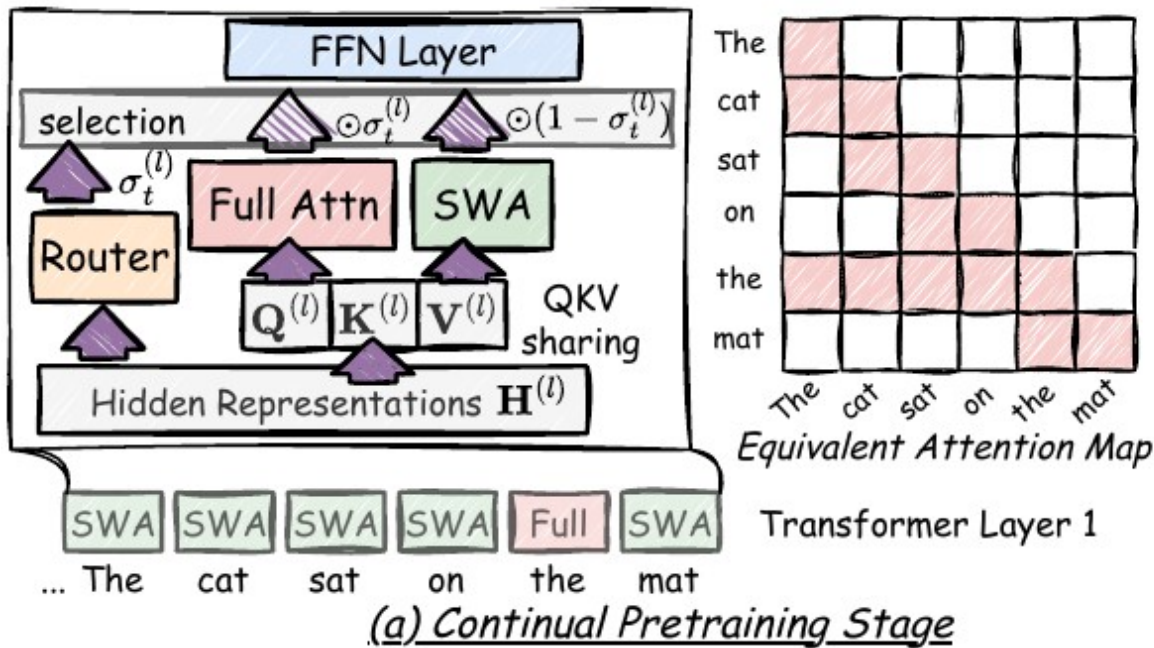
Hard prediction

Different branch outputs

Weaker pressure to use SWA

NLL and branch disagreement are training signals. Inference uses only $R(h)$.

Continual pretraining converts a full-attention model



Train both branches with STE

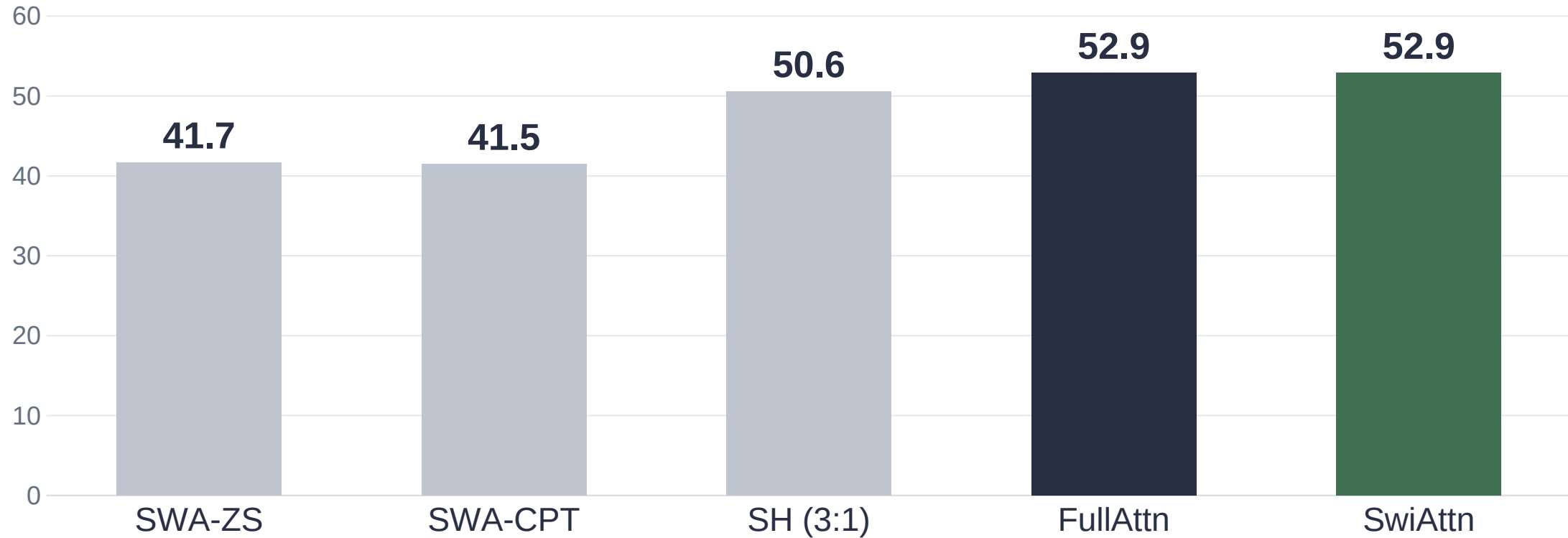
Initialize from a pretrained model
Update the whole hybrid model

Context	Added CPT tokens
4K	20B
32K	8B

1.5B model 22 layers 1,024-token window 23 benchmark datasets

SwiAttn matches FullAttn on in-context retrieval

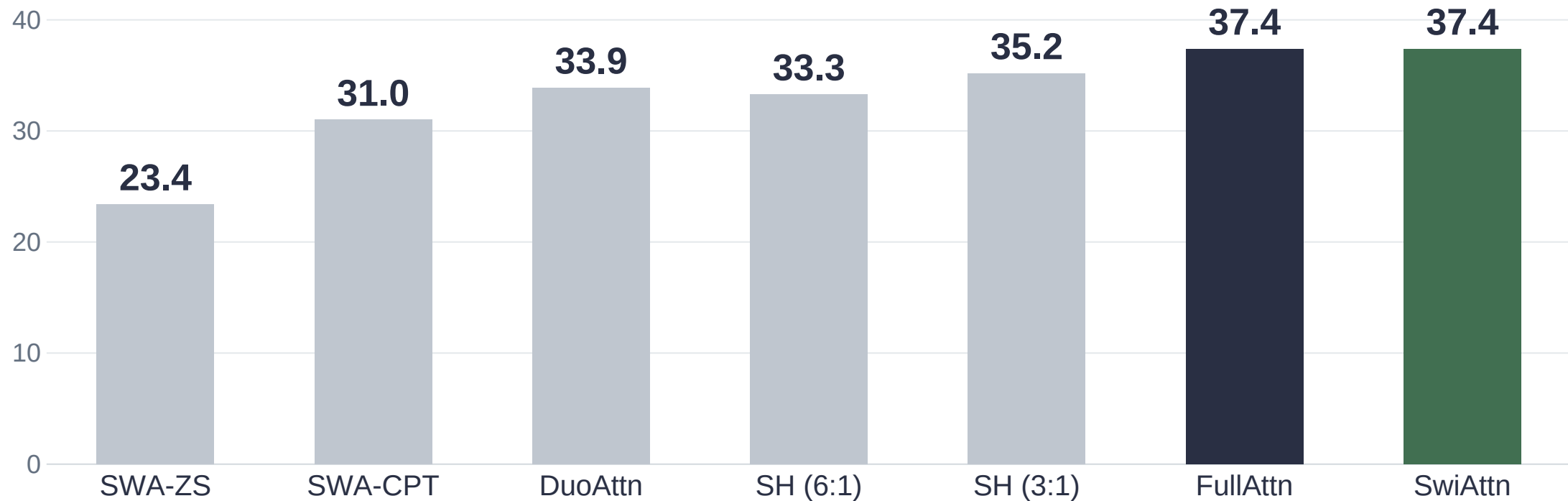
Average score across 6 datasets, up to 4K context



+2.3 points over the static hybrid

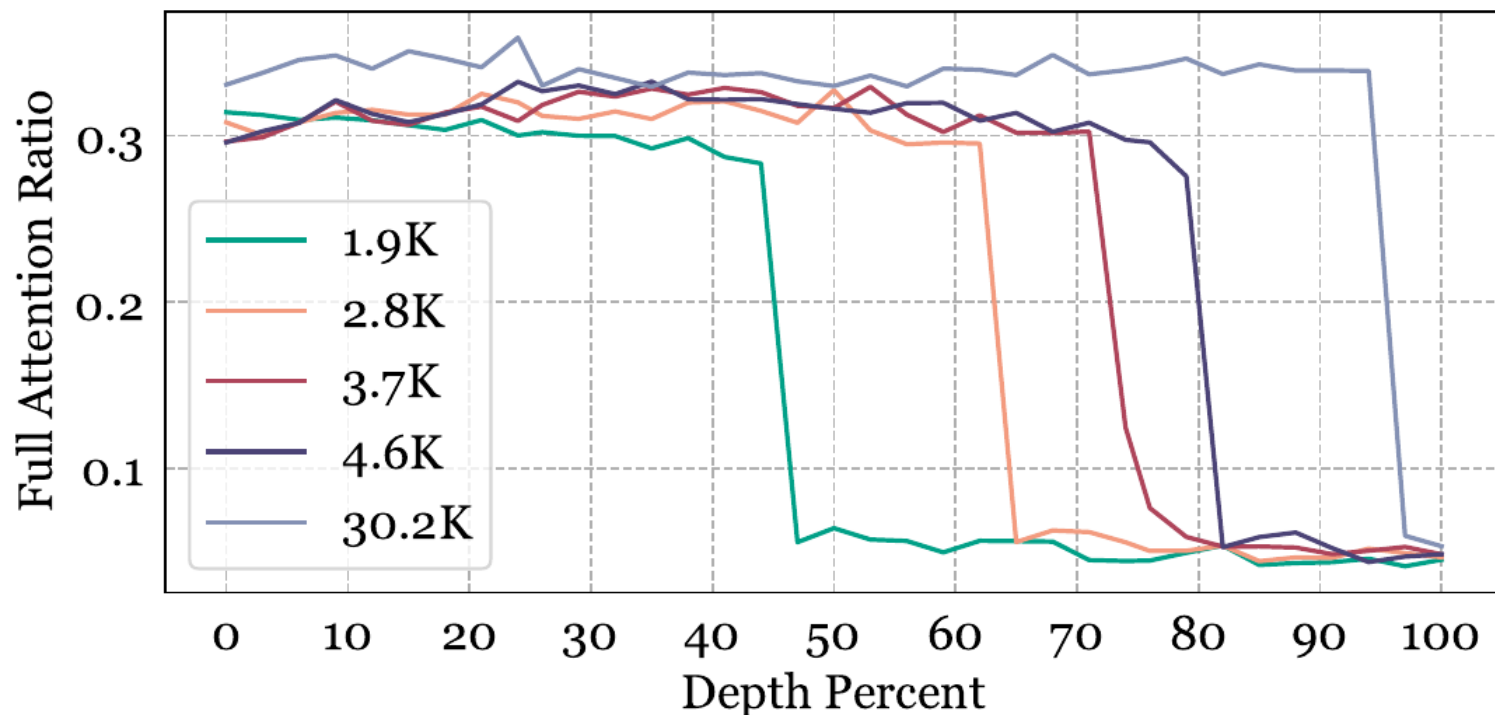
The same LongBench-E average with dynamic routing

Average score across 7 datasets, up to 32K context



+2.2 points over SH (3:1)

Routing responds to where the needed information is



~30%

Full attention when the needle is outside the window

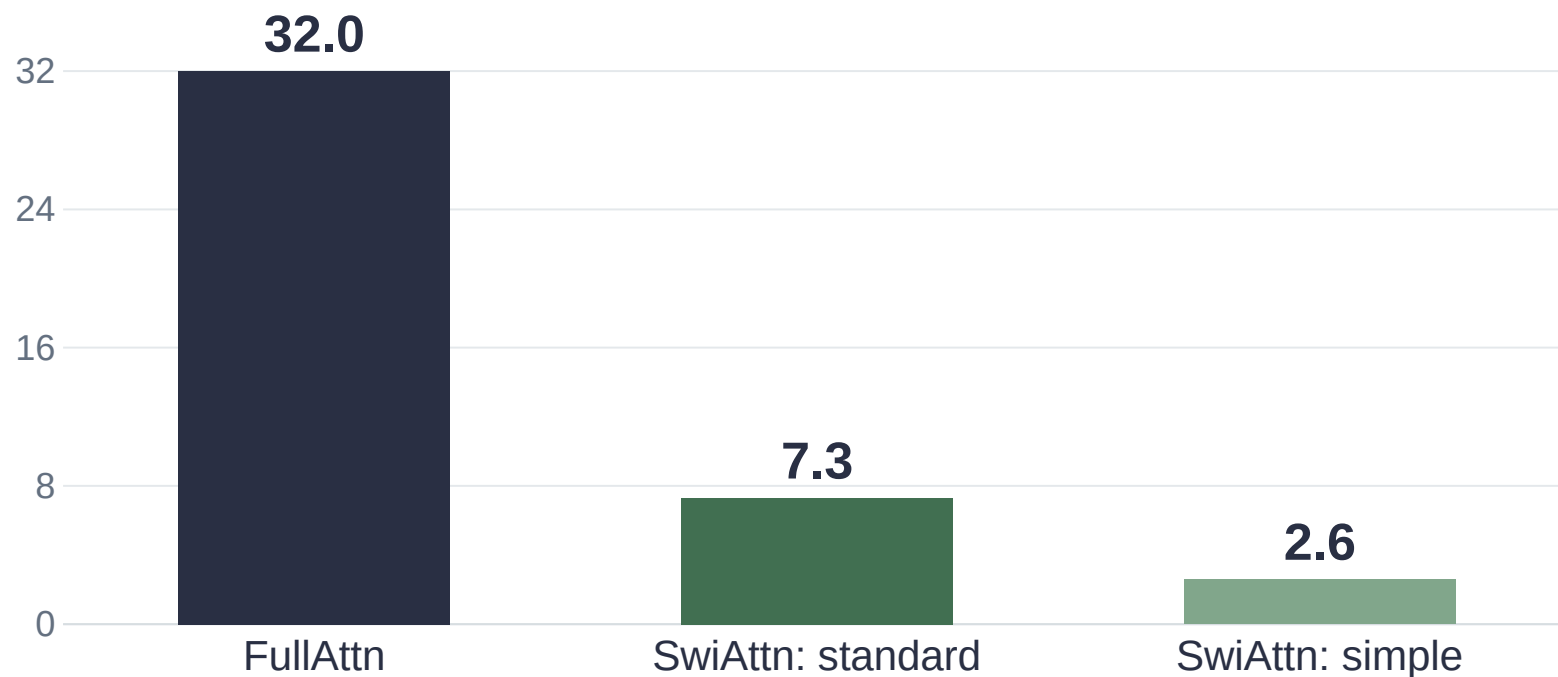
~5%

When the needle is inside the window

100% retrieval accuracy in the tested NIAH grid through 32K context

At 32K, standard decoding reads 4.4× fewer KV tokens

Average KV tokens accessed per layer (thousands), at 32K



4.4×

fewer KV reads
standard distribution

~12×

in the simple setting

KV access reduction. The full cache remains stored. End-to-end speedup is unmeasured.

The savings come with a modest CPT overhead

CPT at 32K

+8.4%

Forward time per layer

+7.9%

Backward time per layer

Practical limits

Full history stays in the KV cache

Savings depend on the input
and the learned routing

Evidence covers models up to 3B

Switch Attention

**Choose full or local attention
for each token at each layer**

FullAttn-level averages on retrieval and LongBench-E
4.4× fewer KV reads in standard 32K decoding

github.com/YushengZhao/SwitchAttentionKernels

Thank you. Questions?

Hard routing and straight-through training

$$p = \text{sigmoid}(R(h))$$

$$g = 1[p > 0.5] + p - \text{stopgrad}(p)$$

$$o = g \cdot o_{\text{full}} + (1 - g) \cdot o_{\text{SWA}}$$

Training: compute both branches and backpropagate through p

Inference: use the hard gate and compute only the chosen branch

Adaptive regularization ablations

Variant	GovReport	MultiNews	SAMSum	Average
Without adaptive weight	24.1	20.7	30.0	24.9
Without NLL	24.3	20.9	30.7	25.3
Without branch MSE	23.9	20.6	29.5	24.7
SwiAttn	25.0	21.3	32.4	26.2

Scores cover three summarization datasets. Compute-matched ablations are not reported.

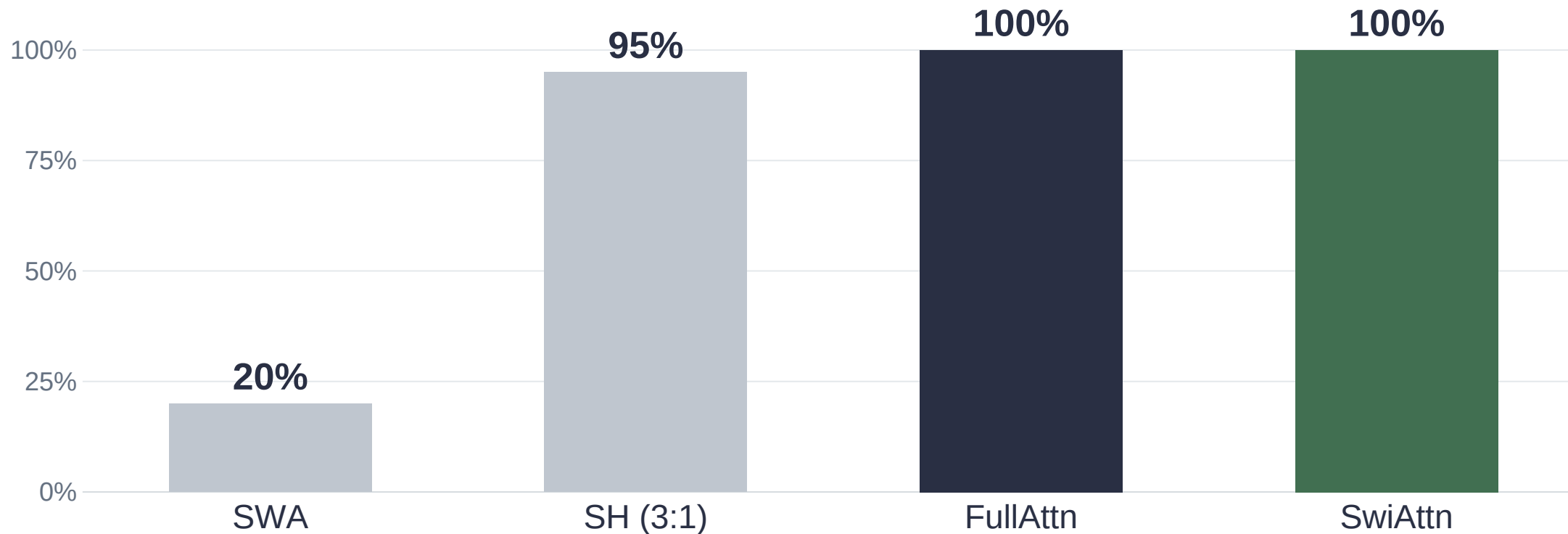
3B results provide a limited scaling check

Evaluation	FullAttn	SwiAttn
Commonsense accuracy	55.9	55.9
In-context retrieval, 6 tasks	55.4	55.8
LongBench-E, 5 tasks	37.2	36.6

The long-context average drops by 0.6 points at 3B

Same training schedule and hyperparameters. Larger industrial scales remain unverified.

NIAH accuracy under the same test configuration



35 context lengths × 35 needle depths × 3 repeats, through 32K

Query tiling makes KV reuse independent of GQA group size

Group routed query positions by branch
Keep each list in position order
Process Bq query positions per tile

Kernel pattern	Dominant arithmetic intensity
Q-outer sparse attention	$G \cdot S / (G + S) \approx G$
SwiAttn query tiling	$Bq \cdot S / (Bq + S) \approx Bq$

G: GQA group size. S: KV span. Approximation when S is large. No latency claim.

LongBench-E results vary by task

Method	GvR	MNs	MFQA	Qas	TQA	SSM	TRec	Avg
SH (3:1)	22.6	20.0	29.6	14.5	68.1	32.3	59.3	35.2
FullAttn	24.9	20.9	33.9	18.6	69.3	30.2	63.7	37.4
SwiAttn	25.0	21.3	35.3	18.0	69.9	32.4	60.0	37.4

Three reported runs: SwiAttn 37.4 ± 0.35 , SH (3:1) 35.2 ± 0.44

$p < 0.01$ vs SH (3:1). A non-significant difference vs FullAttn does not prove equivalence.